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### Temperature indicator

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#### Abstract

A temperature indicator includes a first element, a second element, and circuitry configured to detect whether the first element is in an open circuit condition or a closed circuit condition relative to a contact. Responsive to the indicator being subjected to a temperature exceeding a threshold, the second element changes in a size thereof to cause a change in engagement between the first element and the contact. The circuitry is configured to output a value based on whether the first element and the contact are in the open circuit condition or the closed circuit condition.

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## **Background/Summary**

### **BACKGROUND**

(1) During manufacturing, storage, or transit, many types of objects need to be monitored or tracked due to the temperature sensitivity or fragility of the objects. For example, some types of objects may be susceptible to damage if exposed to certain temperatures (e.g., food or pharmaceutical items). Thus, for quality control purposes and/or the general monitoring of transportation conditions, it is desirable to determine and/or verify the environmental conditions to which the object has been exposed.

### **BRIEF SUMMARY**

(2) According to one aspect of the present disclosure, a device and technique for temperature detection is disclosed. The temperature indicator a first element, a second element, and circuitry configured to detect whether the first element is in an open circuit condition or a closed circuit condition relative to a contact. Responsive to the indicator being subjected to a temperature exceeding a threshold, the second element changes in a size thereof to cause a change in engagement between the first element and the contact. The circuitry is configured to output a value based on whether the first element and the contact are in the open circuit condition or the closed circuit condition.

(3) According to another embodiment of the present disclosure, a temperature indicator includes a housing affixable to an object where the housing has a contact, a first element movable relative to the contact, a second element configured to cause the first element to move relative to the contact, and circuitry configured to detect an engagement condition of the first element relative to the contact. Responsive to the indicator being subjected to a temperature exceeding a threshold, the second element changes in size to thereby cause movement of the first element relative to the contact to change the engagement condition between the first element and the contact, and wherein the circuitry is configured to output a value based on the engagement condition.

(4) According to yet another embodiment of the present disclosure, a temperature indicator includes a contact, a first element movable from a first position to a second position relative to the contact, circuitry coupled to the first element and the contact where the circuitry is configured to detect whether the first element and the contact are in an open circuit condition or a closed circuit condition, and a second element configured to change in a size thereof to cause movement of the first element from the first position to the second position in response to the indicator being subjected to a particular temperature. The first element is prevented from returning from the second position to the first position.

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## **Description**

### **BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS**

(1) For a more complete understanding of the present application, the objects and advantages thereof, reference is now made to the following descriptions taken in conjunction with the

accompanying drawings, in which:

- (2) FIG. 1 is a diagram illustrating an application of an embodiment of a temperature indicator according to the present disclosure;
- (3) FIG. 2 is a block diagram illustrating an embodiment of a temperature indicator according to the present disclosure;
- (4) FIG. 3A is a diagram illustrating an embodiment of a temperature indicator in accordance with the present disclosure in an unactivated state;
- (5) FIG. 3B is a diagram illustrating the temperature indicator of FIG. 3A in accordance with the present disclosure in an activated state;
- (6) FIG. 4A is a diagram illustrating another embodiment of a temperature indicator in accordance with the present disclosure in an unactivated state;
- (7) FIG. 4B is a diagram illustrating the temperature indicator of FIG. 4A in accordance with the present disclosure in an activated state;
- (8) FIG. 5A is a diagram illustrating another embodiment of a temperature indicator in accordance with the present disclosure in an unactivated state; and
- (9) FIG. 5B is a diagram illustrating the temperature indicator of FIG. 5A in accordance with the present disclosure in an activated state.

#### DETAILED DESCRIPTION

(10) Embodiments of the present disclosure provide a device and technique for temperature detection and indication. According to one embodiment, a temperature indicator includes a housing having a temperature detection assembly, switch circuitry, and a radio-frequency identification (RFID) module coupled to the switch circuitry. Responsive to the indicator being subjected to a temperature exceeding a threshold, the temperature detection assembly causes a change in the switch circuitry, wherein the change in the switch circuitry causes a change in a value output by the RFID module when activated. Embodiments of the present disclosure enable temperature event detection using no internal power supply. For example, a temperature-sensing assembly or mechanism closes or opens switch circuitry in response to detecting a particular temperature event. The RFID module can detect the state of the switch circuitry and emit or output a particular value based on whether the switch circuitry is in a closed or open condition. Thus, for example, an RFID reader can be used to activate the RFID module and determine an activation status of the temperature indicator device. Embodiments of the present disclosure provide a temperature indicator that is readily affixable to a container or the like so as to provide at least an indication when a particular container or component has been subjected to a particular environmental temperature. Embodiments of the present disclosure also provide a temperature indicator that is an irreversible, “go-no go” device for indicating that a predetermined temperature has been experienced by the indicator.

(11) With reference now to the Figures and in particular with reference to FIG. 1, exemplary diagrams of a temperature indicator **10** are provided in which illustrative embodiments of the present disclosure may be implemented. FIG. 1 is a diagram illustrating a front view of temperature indicator **10**. In FIG. 1, indicator **10** is a portable device configured to be affixed to or disposed within a transport container containing an object of which temperature events associated therewith are to be monitored. Embodiments of temperature indicator **10** monitor whether an object has been exposed to a particular temperature or environment during manufacturing, storage and/or transport of the object. In some embodiments, temperature indicator **10** may be affixed to a transport container using, for example, adhesive materials, permanent or temporary fasteners, or a variety of different types of attachment devices. The transport container may include a container in which a monitored object is loosely placed or may comprise a container of the monitored object itself. It should be appreciated that FIG. 1 is only exemplary and is not intended to assert or imply any limitation with regard to the environments in which different embodiments may be implemented.

(12) In the embodiment illustrated in FIG. 1, temperature indicator **10** comprises a housing **12**

having a temperature sensing, temperature-sensitive and/or temperature detection assembly **14** disposed therein. In the illustrated embodiment, detection assembly **14** is configured to detect and indicate temperature events relative to indicator **10** (e.g., detecting when indicator **10** (and correspondingly, a container to which indicator **10** is associated with) has been subjected to a particular environmental temperature). In some embodiments, housing **12** is configured and/or constructed from a clear or semi-opaque material having a masking label **20** located on a front side thereof or affixed thereto. In some embodiments, masking label **20** is configured having one or more apertures or “windows” **22** for providing a visual indication of temperature detection. For example, in some embodiments, in response to indicator **10** being subjected to a particular temperature, detection assembly **14** causes a visual indication to be displayed within or through one or more of windows **22** to provide a visual indication that the monitored object has or may have been subjected to some level of temperature. However, it should be understood that other methods may be used to provide a visual indication that detection assembly **14** has moved and/or been otherwise placed into an activated state indicating that indicator **10** has experienced some level of temperature. It should also be understood that housing **12** may be configured and/or manufactured from other materials (e.g., opaque materials having one or more windows **22** formed therein). In some embodiments, housing **12** may be configured without window **22**. For example, as will be described in greater detail below, indicator **10** may be configured to provide visual and/or non-visual indications of whether a temperature condition has been experienced by indicator **10** (e.g., via the use of RFID signals).

(13) FIG. 2 is a block diagram representing and illustrating an embodiment of indicator **10** in accordance with an embodiment of the present disclosure. In some embodiments, temperature indicator **10** may be affixed (permanently or removably) to a printed circuit board and/or otherwise permanently or removably connected to electronic circuitry (e.g., such as a removable cartridge) such that, in response to receipt and/or detection of a temperature condition of a sufficient magnitude and/or exceeding a particular threshold(s), temperature indicator **10** provides an electronic switch closure or opener that may thereby provide an electronic signal/indication of such event. In FIG. 8, indicator **10** includes a switching mechanism **79**, switch circuitry **81**, a wireless communications module **83** coupled to switch circuitry **81**, and temperature detection assembly **14** coupled to switching mechanism **79**. Switching mechanism **79** may be any type of mechanism and/or device used to cause a state change in switch circuitry **81**. For example, in some embodiments, mechanism **79** may comprise a metal switch (e.g., a wire or metallic/conductive strip). In such an embodiment (as will be described in greater detail below), temperature detection assembly **14** may cause a state change in switch circuitry **81** (e.g., changing from an open circuit condition to a closed circuit condition, or vice versa) by causing a change to and/or otherwise actuating/activating switching mechanism **79**. Switch circuitry **81** may comprise one or more switch elements, contacts, and or circuits that are responsive to a change in switching mechanism **79**. Wireless communications module **83** is configured to wirelessly communicate information associated with a state of switch circuitry **81** indicating the activation state of indicator **10** (e.g., based on an open or closed circuit state of circuitry **81**). For example, in one embodiment, wireless communications module **83** includes an RFID module **84**. In some embodiments, RFID module **84** comprises a passive RFID module **84** (e.g., a passive RFID tag) having an RFID integrated circuit or circuitry **86** (e.g., disposed on or as part of a printed circuit board) and a memory **87**, along with an antenna **91**. As a passive RFID module **84**, indicator **10** does not contain a battery (e.g., power is supplied by an RFID reader **100**). For example, when radio waves from reader **100** are encountered by module **84**, antenna **91** forms a magnetic field, thereby providing power to module **84** to energize circuit **86**. Once energized/activated, module **84** may output/transmit information encoded in memory **87**. However, it should be understood that, in some embodiments, RFID module **84** may comprise an active RFID module **84** including a power source (e.g., a battery) that may be configured to continuously, intermittently, and/or according to programmed or event triggers,

broadcast or transmit certain information. It should also be understood that wireless communications module **83** may be configured for other types of wireless communication types, modes, protocols, and/or formats (e.g., short-message services (SMS), wireless data using General Packet Radio Service (GPRS)/3G/4G or through public internet via Wi-Fi, or locally with other radio-communication protocol standards such as Wi-Fi, Z-Wave, ZigBee, Bluetooth®, Bluetooth® low energy (BLE), LoRA, NB-IoT, SigFox, Digital Enhanced Cordless Telecommunications (DECT), or other prevalent technologies). As will be described further below, temperature indicator **10** functions as a temperature fuse such that, in response to encountering a particular temperature (e.g., some temperature exceeding some threshold), an electrically conductive member either opens or closes an electronic switch (e.g., switching mechanism **79**). This configuration enables temperature indicator **10** to be used as a passive temperature sensor/indicator that can be used as part of an electronic signal or circuit. In some embodiments, the temperature sensing capabilities/functions of temperature indicator **10** of the present disclosure needs no power while in the monitoring state. When activated, temperature indicator **10** completes or opens an electrical path of a circuit and thus could be integrated into most any electronic monitoring system.

(14) In the illustrated embodiment, memory **87** includes at least two different stored and/or encoded values **92** and **94**. For example, value **92** may correspond to a value outputted/transmitted by module **84** when switch circuitry **81** is in an open circuit condition or state, and value **94** may correspond to a value outputted/transmitted by module **84** when switch circuitry **81** is in a closed circuit condition or state. As an example, the value **94** may represent an RFID tag identification (ID) number not having an activated temperature switch circuitry **81**, and the RFID tag's ID number may have an additional character (e.g., "0") placed at the end thereof. Value **92** may represent the RFID ID number having an activated temperature switch circuitry **81**, and the RFID tag's ID number may have an additional character at the end thereof being different from the additional character carried by value **94** (e.g., "1"). In the illustrated embodiment, RFID module **84** (e.g., circuitry **86**) is coupled to switch circuitry **81** and can detect whether switch circuitry **81** is in an open or closed circuit condition or state. Thus, for example, switch circuitry **81** may initially be in closed circuit condition or state. Thus, if energized/activated, module **84** would transmit value **94** to reader **100**. If indicator **10** were to be subject to a temperature event (e.g., an environmental temperature exceeding some threshold), mechanism **79** may cause a change in circuitry **81** that would result in circuitry **81** being in an open circuit condition or state. Thus, if now energized/activated (e.g., after the temperature event), module **84** would instead transmit value **92** to reader **100**. Thus, embodiments of the present invention enable indicator **10** to monitor sensitive products/objects to which it is attached for potential damage caused by temperature conditions using electronic indicators (e.g., RFID readers) while indicator **10** does not contain or require any internal power source (e.g., a battery).

(15) The present invention may include computer program instructions at any possible technical detail level of integration (e.g., stored in a computer readable storage medium (or media) (e.g., memory **87**) for causing a processor to carry out aspects of the present invention. Computer readable program instructions described herein can be downloaded to respective computing/processing devices (e.g., communications module **83** and/or RFID module **84**). Computer readable program instructions for carrying out operations of the present invention may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, configuration data for integrated circuitry, or either source code or object code written in any combination of one or more programming languages. In some embodiments, electronic circuitry (e.g., circuitry **86**) including, for example, programmable logic circuitry, field-programmable gate arrays (FPGA), or programmable logic arrays (PLA) may execute the computer readable program instructions by utilizing state information of the computer readable program instructions to personalize the electronic circuitry, in order to perform aspects of the present invention. Aspects of the present

invention are described herein with reference to illustrations and/or block diagrams of methods and/or apparatus according to embodiments of the invention. It will be understood that each block of the illustrations and/or block diagrams, and combinations of blocks in the illustrations and/or block diagrams, may represent a module, segment, or portion of code, can be implemented by computer readable program instructions. These computer readable program instructions may be provided to a processor or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor, create means for implementing the functions/acts specified in the illustrations and/or block diagram block or blocks. These computer readable program instructions may also be stored in a computer readable storage medium that can direct a computing device, a programmable data processing apparatus, and/or other devices to function in a particular manner, such that the computer readable storage medium having instructions stored therein comprises an article of manufacture including instructions which implement aspects of the function/act specified in the illustrations and/or block diagram block or blocks. Switch circuitry **81**, wireless communications module **83**, and/or RFID module **84** may be implemented in any suitable manner using known techniques that may be hardware-based, software-based, or some combination of both. For example, switch circuitry **81**, wireless communications module **83**, and/or RFID module **84** may comprise software, logic and/or executable code for performing various functions as described herein (e.g., residing as software and/or an algorithm running on a processor unit, hardware logic residing in a processor or other type of logic chip, centralized in a single integrated circuit or distributed among different chips in a data processing system). As will be appreciated by one skilled in the art, aspects of the present disclosure may be embodied as a system, method or computer program product. Accordingly, aspects of the present disclosure may take the form of a hardware embodiment, a software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.”

(16) FIG. 3A is a diagram illustrating an embodiment of indicator **10** in accordance with the present disclosure prior to activation (e.g., prior to detecting a particular temperature or time-temperature event), and FIG. 3B is a diagram illustrating an embodiment of indicator **10** in accordance with the present disclosure after activation (e.g., after detecting a particular temperature or time-temperature event). In the embodiment illustrated in FIGS. 3A and 3B, detection assembly **14** includes an absorbent element **120** and a meltable substance **122**. Absorbent element **120** may include any type of material or matter that may absorb substance **122** such that absorbent element **120** swells or increases in size as substance **122** wicks into element **120**. For example, absorbent element **120** may include sponges, hydromorphic materials, expanding plastics available from Industrial Polymer Corporation, and certain cellulosic or fiber-based wicking and swelling materials. Substance **122** may be any type of substance or matter that melts at known or desired time-temperature parameters. For example, substance **122** may include a fatty acid material in the Warm Mark® product from SpotSee of Dallas, Tex.

(17) In the embodiment illustrated in FIGS. 3A and 3B, indicator **10** includes a movable barrier **125**. In FIG. 3A, barrier **125** is located between element **120** and substance **122** such that indicator **10** is in an unarmed state. For example, in some embodiments, substance **122** may be contained within an enclosure or otherwise maintained such that the only migration path for substance **122** if substance **122** melts is toward element **120**. Thus, in such an embodiment, if indicator **10** is subject to a temperature that would otherwise activate indicator **10** before indicator **10** is armed, substance **122** would be prevented from contacting and/or wicking into element **120**. In FIG. 3B, barrier **125** has been moved and/or withdrawn from being between element **120** and substance **122**, thereby arming indicator **10**, such that substance **122** is in contact with and/or in close proximity to element **120**. Thus, in the armed state, if indicator **10** is subject to a temperature that would activate indicator **10**, substance **122** can migrate to and/or into absorbent element **120** should certain time-



temperature parameters be met. In operation, before meltable substance **122** has been exposed to certain time-temperature parameters, meltable substance **122** remains in a solid state and does not migrate onto or into absorbent element **120** (even with barrier **125** removed from between them). (18) In the embodiment illustrated in FIGS. **3A** and **3B**, switch circuitry **81** includes switch elements **130**, **132**, **134**, **136**, **138**, and **140** each coupled or connected to RFID module **84**. In the illustrated embodiment, switching mechanism **79** includes contacts **152**, **154**, and **156**. In the embodiment illustrated in FIGS. **3A** and **3B**, contacts **152**, **154**, and **156** are spaced apart from each other and are spaced apart from substance **122** by varying distances. In FIG. **3A**, in the unactivated state of indicator **10**, elements **130** and **132** are connected to contact **152**, elements **134** and **136** are connected to contact **154**, and elements **138** and **140** are connected to contact **156**. In this particular embodiment, switch circuitry **81** includes three separate circuits **81.sub.1**, **81.sub.2**, and **81.sub.3** each in a closed state or closed circuit condition (e.g., elements **130** and **132** and contact **152** forming a first circuit **81.sub.1**; elements **134** and **136** and contact **154** forming a second circuit **81.sub.2**; and elements **138** and **140** and contact **156** forming a third circuit **81.sub.3**). It should be understood that the number of different circuits forming switch circuitry **81** may vary. Switch elements **130**, **132**, **134**, **136**, **138**, and **140**, and contacts **152**, **154**, and **156**, may comprise electrically conductive wires, traces, pads, posts, and/or other type of electronic conductive components.

(19) In FIG. **3B**, indicator **10** has been subjected to a time and temperature parameter such that substance **122** (all or a portion thereof) has melted, wicked into, and been absorbed by element **120**, thereby causing element **120** to swell and/or increase in size (in the direction(s) indicated by arrow **160**). As illustrated in FIG. **3B**, the time period in which indicator **10** has been exposed to a certain temperature may result in varying degrees of absorption of substance **122** by element **120** such that the amount of swelling of element **120** may vary over time over a length of element **120** (e.g., greater swelling near contact **156** (e.g., closer to an original location of substance **122**) than near contact **152**). In the illustrated embodiment, the time and temperature parameter has been enough to cause an open circuit condition as to switch elements **134**, **136**, **138**, and **140** while a closed circuit condition still exists as to switch elements **130** and **132**. For example, at a particular temperature, substance **122** begins to melt and substance **122** begins to wick into and be absorbed by element **120**. As element **120** absorbs a portion of substance **122** over some time period or parameter, element **120** begins to change in size (e.g., element **120** swells), thereby initially causing a break or disconnection between elements **138** and/or **140** with contact **156** (e.g., an open circuit condition in the first circuit **81.sub.1**). After an additional time period or parameter, element **120** absorbs an additional amount of substance **122** thereby causing additional swelling of element **120** toward contact **154** such that the swelling of element **120** causes a break or disconnection between elements **134** and/or **136** with contact **154** (e.g., an open circuit condition in the second circuit **81.sub.2**). In the illustrated embodiment, a time and temperature parameter has not yet reached a value where the third circuit **81.sub.3** has changed from a closed circuit state to an open circuit state. However, as the time and temperature parameter continues (e.g., some temperature is exceeded for some period of time), all three circuits (e.g., circuits **81.sub.1**, **81.sub.2**, and **81.sub.3**) would change from a closed state to an open state.

(20) Thus, in operation, RFID module **84** detects the open circuit condition of circuitry **81** and outputs or transmits certain values based on such conditions. For example, in this embodiment, RFID module **84** may output a separate value for each of circuits **81.sub.1**, **81.sub.2**, and **81.sub.3**, thereby enabling a determination of a time period in which indicator **10** may have been subjected to some temperature parameter. For example, at a particular moment in time, circuit **81.sub.1** may be open while circuits **81.sub.2** and **81.sub.3** remain closed. At this instant, RFID module **84** may output a value for circuit **81.sub.1** indicating its open state while outputting different values for circuits **81.sub.2** and **81.sub.3** indicating their closed states. However, after an additional time period of being subjected to a particular temperature parameter, element **120** may have expanded or

swelled an additional amount by absorbing an additional amount of substance **122** such that both circuits **81.sub.1** and **81.sub.2** are open while circuit **81.sub.3** remains closed. At this later instant in time, RFID module **84** may output a value for circuits **81.sub.1** and **81.sub.2** indicating their open state while outputting a different value for circuit **81.sub.3** indicating its closed state. Thus, embodiments of the present invention enable the indication and/or determination of a time parameter associated with a particular temperature parameter of exposure. RFID module **84** may be configured to cycle through different output values where each value corresponds to a different circuit **81.sub.1**, **81.sub.2**, and **81.sub.3** (e.g., outputting value **92** or **94** corresponding to each circuit **81.sub.1**, **81.sub.2**, and **81.sub.3**); however, it should be understood that RFID module **84** may be otherwise configured to indicate different conditions corresponding to each circuit **81.sub.1**, **81.sub.2**, and **81.sub.3** (e.g., outputting a three-digit value where each digit represents the closed or open state of a respective circuit **81.sub.1**, **81.sub.2**, and **813**; outputting unique alternate values for each circuit **81.sub.1**, **81.sub.2**, and **81.sub.3**; outputting one of four unique values where a first value would represent having an unactivated indicator **10** (e.g., no time-temperature excursion having been experienced), a second value represents a change in circuit **81i**, a third value represents a change in each of circuits **81.sub.1** and **81.sub.2**, and a fourth value represents a change in each of circuits **81.sub.1**, **81.sub.2**, and **81.sub.3**). Thus, in operation, when the RFID module **84** is queried, module **84** checks the condition of circuits **81.sub.1**, **81.sub.2**, and **81.sub.3**) and outputs the appropriate or corresponding value(s). Software or code in reader **100** or another application could then parse the value(s) and make a determination as to whether (or to what extent) a time-temperature event has occurred.

(21) In the illustrated embodiment, the expanding or swelling of element **120** causes a circuit connection to be broken or separated (e.g., between elements **138** and/or **140** and contact **156**). However, it should be understood that indicator **10** may be otherwise configured. For example, in some embodiments, contacts **152**, **154**, and **156** may be configured from a frangible conductive material that breaks or separates into one or more pieces as a result of the expansion or swelling of absorbent element **120**, thereby causing circuitry **81** (e.g., circuits **81.sub.1**, **81.sub.2**, and **81.sub.3**) to change from a closed state to an open state. In another embodiment, substance **122** may have conductive properties (e.g. infused conductive particles) that close openings in conductive paths. In this embodiment, circuits **81.sub.1**, **81.sub.2**, and **81.sub.3** may initially be in an open state (e.g., in an unactivated state of indicator **10**) such that as substance **122** wicks into element **120** and element **120** expands/swells, element **120** (with substance **122**) engages various switching elements **130**, **132**, **134**, **136**, **138**, and/or **140** to close one or more circuits **81.sub.1**, **81.sub.2**, and **81.sub.3**. In yet another embodiment, switching elements **130**, **132**, **134**, **136**, **138**, and **140** may initially be spaced apart from respective contacts **152**, **154**, and **156** (e.g., in an unactivated state of indicator **10**) such that as substance **122** wicks into element **120** and element **120** expands/swells, various switching elements **130**, **132**, **134**, **136**, **138**, and/or **140** move into engagement with respective contacts **152**, **154**, and **156**, thereby changing one or more circuits **81.sub.1**, **81.sub.2**, and **81.sub.3** from an open state to a closed state (e.g., gaps between switching elements **130**, **132**, **134**, **136**, **138**, and **140** and respective contacts **152**, **154**, and **156** as a result of the swelling of element **120**).

(22) FIGS. **4A** and **4B** are diagrams illustrating another embodiment of temperature indicator **10** in accordance with the present disclosure. FIG. **4A** is a diagram illustrating an embodiment of indicator **10** in accordance with the present disclosure prior to activation (e.g., prior to detecting a particular temperature or time-temperature event), and FIG. **4B** is a diagram illustrating an embodiment of indicator **10** in accordance with the present disclosure after activation (e.g., after detecting a particular temperature or time-temperature event). In the embodiment illustrated in FIGS. **4A** and **4B**, detection assembly **14** includes absorbent element **120** and meltable substance **122**. In the illustrated embodiment, switching mechanism **79** includes a contact member **200** disposed against and/or adjacent to absorbent element **120**. In some embodiments, contact member **200** is a printed circuit board (PCB) having a conductive contact **202** in the form of a via **204**. For

example, contact **202** may include via **204** extending through member **200** and coated with a conductive material. In the illustrated embodiment, RFID module **84** is coupled to member **200** such that member **200** is sandwiched between RFID module **84** and element **120**; however, it should be understood that RFID module **84** may be otherwise located.

(23) In the embodiment illustrated in FIGS. **4A** and **4B**, switch circuitry **81** includes switch elements **210** and **212**. Switch element **212** is coupled to RFID module **84** and contact **202** (e.g., one end of element **212** is coupled to RFID module **84** and another end of switch element **212** is coupled to and/or in engagement with contact **202**). Switch element **210** is also coupled to RFID module **84**. In FIG. **4A** (e.g., an unactivated state of indicator **10**), element **210** extends around element **120** and extends into and through a via **220** formed in element **120** to engage contact **202** (e.g., one end of switch element **210** coupled to RFID module **84** and another end of switch element **210** extending through via **220** and into engagement with contact **202**). For example, in the illustrated embodiment, switch element **212** engages contact **202** from a side or surface **222** of member **200**. An opposite side or surface **224** of member **200** (e.g., opposite side **222**) is disposed against and/or adjacent a side or surface **226** of element **120**. On an opposite side or surface **228** of element **120** (e.g., opposite side **226**) are holding members **230** and **232**. Holding members **230** and **232** may be tape strips, mounted PCB elements, straps, fasteners, or any other suitable material to retain a portion of switch element **210** adjacent and/or near side **228** of element **120**. As illustrated in FIG. **4A**, switch elements **210** and **212** engage contact **202** from opposite sides **224** and **222**, respectively, of member **200** such that element **120** is disposed between a portion **240** of switch element **210** being held by members **230** and **232** and member **200**. Switch elements **210** and **212** and contact **202** may comprise electrically conductive wires, traces, pads, posts, coatings and/or other type of electronic conductive components.

(24) In the embodiment illustrated in FIG. **4A**, a distal end **250** of switch element **210** (e.g., an end opposite an end of element **210** coupled to RFID module **84**) is biased in a direction indicated by arrow **252** to enable insertion of end **250** into via **204** and into engagement with contact **202**. For example, in the illustrated embodiment, vias **204** and **220** may be misaligned or offset from each other (e.g., at least a portion thereof) such that in a free or unbiased state, end **250** is not aligned with via **204**. Thus, end **250** of switch element **210** functions similar to a spring element having a mechanical memory such that if end **250** is withdrawn from via **204** (e.g., towards side **228**), end **250** returns to an unbiased or free state/position that is misaligned with via **204** (e.g., as depicted in FIG. **4B**). As illustrated in FIG. **4A**, the biasing of end **250** also ensures engagement between end **250** and contact **202**.

(25) Thus, in operation, in an unactivated state (e.g., prior to a time-temperature parameter that would cause substance **122** to melt as depicted in FIG. **4A**), switch circuitry **81** is in a closed circuit state or condition as switch elements **210** and **212** are both in engagement with contact **202**. In this state, when queried, RFID module **84** transmits value **94** indicating a closed circuit condition, thereby indicating that indicator **10** has yet to be exposed to a time-temperature parameter that would otherwise activate indicator **10**. In response to indicator **10** being subjected to some time-temperature parameter that causes substance **122** to melt, substance wicks into and is otherwise absorbed by element **120**, thereby causing element **120** to expand or swell (e.g., in the direction(s) indicated by arrow **260**). For example, before substance **122** has been exposed to certain time-temperature parameters, substance **122** remains in a solid state and does not migrate into element **120**. In some embodiments, a removable arming barrier (e.g., such as barrier **125** depicted in FIGS. **3A** and **3B**) may be located between substance **122** and element **120** that can be removed to arm indicator **10**, thereby enabling indicator **10** to be field activated such that exposures to high temperatures before a time period of interest does not result in substance **122** melting and wicking into element **120**.

(26) As illustrated in FIG. **4B**, in response to indicator **10** being subjected to some time-temperature parameter that exceeds some threshold that causes substance **122** to melt, thereby also causing

element **120** to expand or swell as substance **122** wicks into and is absorbed by element **120**, the swelling of element **120** causes end **250** of element **210** to withdraw from via **204** and disengage from contact **202**. As a result, a lack of conductive engagement between end **250** of switch element **210** and contact **202** causes circuitry **81** to change from a closed circuit to an open circuit. In this state, when queried, RFID module **84** transmits value **92** indicating an open circuit condition, thereby indicating that indicator **10** has been exposed to a time-temperature parameter that activated indicator **10**. In some embodiments, switch element **210** may include additional material, length, or “slack” between RFID module **84** and holding members **230** and **232** to enable the expansion of element **120**; however, it should be understood that the connection of switch element **210** with RFID module **84** may be otherwise configured. Additionally, in the embodiment illustrated in FIG. 4B, for ease of illustration and description, element **120** depicts a uniform swelling or expansion over a length of element **120** (e.g., from an end near an initial location of substance **122** to a distal end thereof); however, it should be understood that element **120** change in size in a non-uniform manner over its length as a result of a time parameter in which indicator is subjected to a temperature parameter causing substance **122** to melt.

(27) As illustrated in FIG. 4A, after end **250** of switch element **210** has withdrawn from via **204**, end **250** is free to return to an unbiased state (e.g., moving in a direction indicated by arrow **270**) due to the stiffness or mechanical shape memory of switch element **210**. For example, as element **120** swells, end **250** of switch element **210** moves away from member **200** in a direction along an axis indicated by arrow **260** and, once end **250** exits via **202**, end **250** moves in the direction of arrow **270** to its initial unbiased state. In an unbiased state, end **250** of switch element **210** is misaligned with via **202**, thereby preventing end **250** from re-entering via **202** should element **120** decrease in size or contract as a result of substance **122** evaporation or otherwise. Thus, embodiments of the present invention provide a non-reversible time-temperature indicator **10**.

(28) In the embodiment illustrated in FIGS. 4A and 4B, a single switch element **210** (along with switch element **212** and contact **202**) forming a single circuit is depicted; however, it should be understood that indicator **10** could include additional circuits (e.g., similar to circuits **81.sub.1**, **81.sub.2**, and **81.sub.3** depicted and described in connection with FIGS. 3A and 3B). For example, circuitry similar to element **210** (along with switch element **212** and contact **202**) may be replicated and located at varying spacing or intervals along element **120** relative to an initial location of substance **122** such that a time-temperature parameter of exposure may cause different circuits to be closed or open, thereby enabling a time parameter determination associated with the event.

(29) FIGS. 5A and 5B are diagrams illustrating another embodiment of temperature indicator **10** in accordance with the present disclosure. FIG. 5A is a diagram illustrating an embodiment of indicator **10** in accordance with the present disclosure prior to activation (e.g., prior to detecting a particular temperature or time-temperature event), and FIG. 5B is a diagram illustrating an embodiment of indicator **10** in accordance with the present disclosure after activation (e.g., after detecting a particular temperature or time-temperature event). The embodiment of indicator **10** depicted in FIGS. 5A and 5B is similar in most respects to indicator **10** depicted and described in connection with FIGS. 4A and 4B except that in FIGS. 4A and 4B, circuitry **81** is initially in a closed circuit condition in an unactivated state of indicator **10** and in an open circuit condition in an activated state on indicator **10**. In FIGS. 5A and 5B, an embodiment of indicator **10** is depicted where circuitry **81** is initially in an open circuit condition in an unactivated state of indicator **10** and in a closed circuit condition in an activated state on indicator **10**. For example, in the embodiment illustrated in FIGS. 5A and 5B, end **250** of switch element **210** includes a non-conductive shielding **300** extending along a portion thereof prior to reaching a distal location of end **250**. At the distal location of end **250**, there is a lack of shielding **300** and/or conductive segment **302**.

(30) Thus, as illustrated in FIG. 5A, in an unactivated state of indicator **10**, end **250** of switch element **210** is disposed within via **204** and extends at least partially therethrough such that shielding **300** is disposed against contact **202** within via **204** and segment **302** is located beyond or

outside via **204**, thereby resulting in segment **302** disposed in a spaced apart relation from contact **202** and in a lack of conductive contact with contact **202**. However, referring to FIG. 5B, in response to a time-temperature parameter activating indicator **10**, thereby causing substance **122** to melt and element **120** to expand/swell, end **250** of switch element **210** is drawn through via **204** in the direction indicated by arrow **304** relative to member **200** such that conductive segment **302** engages contact **202**, thereby resulting in a closed circuit condition of circuitry **81**. Thus, it should be understood that embodiments of indicator **10** may be configured to initially be in either a closed circuit or open circuit state that changes based on an activation of indicator **10**.

(31) Thus, embodiments of the present disclosure enable temperature and/or time-temperature event detection using a temperature indicator having a small footprint using a temperature-sensitive assembly with a passive RFID tag that gives a different reading depending upon the status of the temperature switch circuitry. Because the RFID tag is passive, the temperature indicator does not need a battery or other external power source. Further, the configuration of the temperature indicator enables the temperature indicator to be irreversible once activated (or subjected to a sufficient magnitude and/or time of temperature event). Additionally, the temperature indicator of the present disclosure may be configured with a single or multiple indication mechanisms (e.g., with window **22** where indicator **10** provides a visual indication of activation, with a combination of window **22** and with an RFID transmitted value indicating the activation status, or with an RFID transmitted value without window **22**).

(32) The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(33) The corresponding structures, materials, acts, and equivalents of all means or step plus function elements in the claims below are intended to include any structure, material, or act for performing the function in combination with other claimed elements as specifically claimed. The description of the present disclosure has been presented for purposes of illustration and description, but is not intended to be exhaustive or limited to the disclosure in the form disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the disclosure. The embodiment was chosen and described in order to best explain the principles of the disclosure and the practical application, and to enable others of ordinary skill in the art to understand the disclosure for various embodiments with various modifications as are suited to the particular use contemplated.

## Claims

1. A temperature indicator, comprising: a housing having a detection assembly, the detection assembly including: first and second elements; and circuitry configured to detect whether the first element is in an open circuit condition or a closed circuit condition relative to a contact; and wherein responsive to the indicator being subjected to a temperature exceeding a threshold, the second element changes in a size thereof to cause a change in engagement between the first element and the contact, and wherein the circuitry is configured to output a value based on whether the first element and the contact are in the open circuit condition or the closed circuit condition.
2. The temperature indicator of claim 1, wherein the temperature detection assembly includes a substance configured to melt in response to the temperature exceeding the threshold.
3. The temperature indicator of claim 2, wherein the second element is configured to absorb at least a portion of the melted substance.

4. The temperature indicator of claim 1, wherein the circuitry is configured to output a first value in the open circuit condition and a second value in the closed circuit condition.
5. The temperature indicator of claim 1, further comprising a barrier disposed adjacent the second element, the barrier removable from being adjacent the second element to enable the second element to respond to the temperature exceeding the threshold.
6. The temperature indicator of claim 2, further comprising a barrier disposed between the second element and the substance, the barrier removable from being between the second element and the substance to enable the second element to respond to the temperature exceeding the threshold.
7. The temperature indicator of claim 1, wherein the change in the engagement results from the first element moving from a first position to a second position, and wherein the detection assembly is configured to prevent the first element from returning to the first position from the second position.
8. A temperature indicator, comprising: a housing affixable to an object, the housing having: a contact; a first element movable relative to the contact; a second element configured to cause the first element to move relative to the contact; and circuitry configured to detect an engagement condition of the first element relative to the contact; and wherein responsive to the indicator being subjected to a temperature exceeding a threshold, the second element changes in size to thereby cause movement of the first element relative to the contact to change the engagement condition between the first element and the contact, and wherein the circuitry is configured to output a value based on the engagement condition.
9. The temperature indicator of claim 8, wherein the contact and the first element are electrically conductive.
10. The temperature indicator of claim 8, wherein the circuitry is configured to output a first value when the first element is in engagement with the contact and a output a second value when the first element is disengaged from the contact.
11. The temperature indicator of claim 8, wherein the second element is configured to absorb at least a portion of a substance to change the size.
12. The temperature indicator of claim 8, wherein the change in the engagement condition results from the first element moving from a first position to a second position, and wherein the first element is prevented from returning to the first position from the second position.
13. A temperature indicator, comprising: a contact; a first element movable from a first position to a second position relative to the contact; circuitry coupled to the first element and the contact, the circuitry configured to detect whether the first element and the contact are in an open circuit condition or a closed circuit condition; and a second element configured to change in a size thereof to cause movement of the first element from the first position to the second position in response to the indicator being subjected to a particular temperature, the first element prevented from returning from the second position to the first position.
14. The temperature indicator of claim 13, wherein the second element is configured to absorb at least a portion of a substance to change the size.
15. The temperature indicator of claim 14, further comprising a barrier disposed between the second element and the substance, the barrier removable from being between the second element and the substance to enable the second element to respond to the particular temperature.
16. The temperature indicator of claim 13, wherein the second element is an absorbent element.
17. The temperature indicator of claim 14, further comprising a barrier element disposed between the substance and the second element to prevent migration of the substance into the second element.
18. The temperature indicator of claim 13, wherein the circuitry is configured to output an indication of the open circuit condition.
19. The temperature indicator of claim 13, wherein the first element is configured to maintain either the open circuit condition or the closed circuit condition in response to the movement.

20. The temperature indicator of claim 13, wherein the circuitry is configured to output an indication of the closed circuit condition.

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